Skills Summary

Multistep Surface Area and Scale Design

Use this reference while completing your worksheet or when studying.

Method for every problem here: (1) name the surface or region, (2) break it into faces or rectangles you have a formula for, (3) decide which pieces are actually included, (4) add, and keep the unit — lengths in m or cm, areas in m^2 or cm^2.

1. Surface area of a rectangular prism

Formula: $SA = 2(\ell w + \ell h + wh)$

The six faces come in three matching pairs, which is where the 2 comes from. If a face is missing (an open box, an unpainted floor), do *not* use this formula — add the faces you actually need, one pair at a time.

Example: A closed box is 10 cm by 4 cm by 3 cm. Find its surface area.
Step 1. All six faces are wanted, so the paired-faces formula applies directly.
Step 2. $SA = 2(10 \cdot 4 + 10 \cdot 3 + 4 \cdot 3) = 2(40 + 30 + 12) = 2(82) \Rightarrow$ **164** cm^2

Try it: A closed box is 9 cm by 5 cm by 2 cm. Find its surface area.

check: **146** cm^2

2. Surface area of a cylinder (surface-area-cylinder)

Formulas: curved side $= 2\pi rh$ (a rectangle $2\pi r$ long and h tall)
one circular end $= \pi r^2$ closed can: $SA = 2\pi rh + 2\pi r^2 = 2\pi r(r + h)$

Count the ends the problem really has: a label has none, an open tub has one, a sealed can has two.

Example: A closed cylinder has radius 5 cm and height 10 cm. Find its surface area.
Step 1. Sealed at both ends, so the surface is the curved side plus two circles: $2\pi r(r + h)$.
Step 2. $2\pi(5)(5 + 10) = 150\pi \approx 471.2388\ldots \Rightarrow$ **471.24** cm^2

Try it: A closed cylinder has radius 2 cm and height 7 cm. Find its surface area.

check: **113.10** cm^2

3. Composite plan areas (composite-area)

Two ways, same answer:

Add: cut the shape into rectangles, find each area, add them. **Subtract:** enclose it in one big rectangle and take away the missing corner.

Pick whichever needs fewer measurements, then check with the other.

Example: A patio is a 9 m by 5 m rectangle with a 3 m by 2 m corner cut out. Find its area.
Step 1. The shape is one rectangle with a bite taken out, so subtracting is quicker than cutting it into pieces.
Step 2. $9 \cdot 5 - 3 \cdot 2 = 45 - 6 \Rightarrow$ **39** $\mathbf{m^2}$

Try it: A deck is a 12 m by 7 m rectangle with a 4 m by 3 m corner cut out. Find its area.

check: **72** $\mathbf{m^2}$

4. Scale factors: length, area, volume

The rule that catches everyone: if every *length* is multiplied by k, then every *area* is multiplied by k^2 and every *volume* by k^3.

Safest route: convert the lengths first, then use the area formula on the real lengths. If you start from a drawing's area instead, remember to multiply by k^2, not k.

Example: A drawing uses 1 cm = 3 m. A room is 4 cm by 2.5 cm on the drawing. Find the real area.
Step 1. Area is asked for, so convert each length with the scale first and multiply afterwards.
Step 2. Real lengths: $4 \cdot 3 = 12$ m and $2.5 \cdot 3 = 7.5$ m, so the area is $12 \cdot 7.5$. $\Rightarrow$ **90** $\mathbf{m^2}$
(Same as drawing area 10 cm^2 times 3^2.)

Try it: A drawing uses 1 cm = 6 m. A room is 3 cm by 2 cm on the drawing. Find the real area.

check: **216** $\mathbf{m^2}$

(!) Watch out: Three habits fix most lost marks here — count only the faces that exist (an unpainted floor is not part of the answer), never apply a length scale factor straight to an area, and finish with the right unit: m^2 for area, m for a length.